ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

PHYSICAL SCIENCES

PAPER 1 (PHYSICS)

MARKS: 150 **TIME: 2 hours**

This question paper consists of 6 pages, including a data sheet.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Number your answers correctly according to the numbering in this question paper.
2. You may use a non-programmable calculator and appropriate mathematical instruments.
3. Show ALL formulae and substitutions in ALL calculations. Round off final numerical answers to a minimum of TWO decimal places.
4. Give brief motivations, discussions, et cetera where required. Use the data sheet below.

DATA SHEET

Acceleration due to gravity	g = 9,8 m·s^{-2}	Speed of light in a vacuum	c = 3,0 × 10^{8} m·s^{-1}
Planck's constant	h = 6,63 × 10^{-34} J·s	Charge on an electron	e = −1,6 × 10^{-19} C

$v_f = v_i + a\Delta t$	$\Delta x = v_i\Delta t + \frac{1}{2}a\Delta t^2$	$v_f^2 = v_i^2 + 2a\Delta x$	$\Delta x = \frac{v_i + v_f}{2}\Delta t$
$E_P = mgh$	$E_K = \frac{1}{2}mv^2$	$E_M = E_K + E_P$	$v = f\lambda \quad T = \frac{1}{f}$
$E = hf \quad c = f\lambda$	$Q = nq_e$	$I = \frac{Q}{\Delta t} \quad R = \frac{V}{I}$	$R_s = R_1 + R_2 \quad \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$

QUESTION 1: MULTIPLE-CHOICE QUESTIONS

Four options are given as possible answers. Choose the answer and write only the letter (A–D) next to the question number.

1.1 Which ONE of the following is a VECTOR quantity? (2)

A Speed B Distance
C Displacement D Time

1.2 An object moves in a straight line at a constant velocity. Its acceleration is … (2)

A zero. B constant, but not zero.
C increasing. D decreasing.

1.3 The gradient of a position-time graph represents the object's … (2)

A acceleration. B velocity.
C displacement. D distance.

1.4 A ball falls freely with no air resistance. Which quantity stays CONSTANT? (2)

A Kinetic energy B Gravitational potential energy
C Mechanical energy D Speed

1.5 The SI unit of frequency is the … (2)

A second. B metre per second.
C hertz. D metre.

1.6 Sound CANNOT travel through … (2)

A water. B steel.
C air. D a vacuum.

1.7 Which type of electromagnetic radiation has photons with the HIGHEST energy? (2)

A Radio waves B Visible light
C Ultraviolet D Gamma rays

1.8 Two identical metal spheres carrying charges of +6 nC and −2 nC touch and are then separated. The charge on EACH sphere is now … (2)

A +4 nC B +2 nC
C −2 nC D +8 nC

1.9 When resistors are connected in PARALLEL, the effective resistance is … (2)

A greater than the largest resistor. B smaller than the smallest resistor.
C equal to the sum of the resistors. D equal to the average of the resistors.

1.10 Outside a bar magnet, magnetic field lines point from the … (2)

A north pole to the south pole. B south pole to the north pole.
C north pole to the north pole. D south pole to the south pole.

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QUESTION 2

A learner walks 300 m east and then 400 m north. The whole walk takes 500 s.

2.1 Define the term *vector*. (2)

2.2 Define the term *displacement*. (2)

2.3 Is distance a scalar or a vector quantity? (1)

2.4 Write down the total distance walked. (1)

2.5 Calculate the magnitude of the learner's resultant displacement. (3)

2.6 Calculate the direction of the resultant displacement, measured from east. (2)

2.7 Calculate the learner's average speed. (3)

2.8 Calculate the learner's average velocity. (3)

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QUESTION 3

A car starts from rest and accelerates uniformly at 2,5 $m \cdot s^{-2}$ for 8 s. It then moves at a constant velocity for 10 s, and then brakes uniformly to come to rest in 5 s. The car moves in a straight line.

3.1 Define the term *acceleration*. (2)

3.2 Calculate the velocity of the car after 8 s. (3)

3.3 Calculate the displacement of the car during the first 8 s. (3)

3.4 Calculate the acceleration of the car while it brakes. (3)

3.5 Sketch the velocity-time graph for the whole 23 s. Show the values of velocity and time on your graph. (4)

3.6 Use the graph to calculate the total displacement of the car. (4)

3.7 Calculate the average velocity of the car for the whole trip. (3)

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QUESTION 4

A ball of mass 0,5 kg is dropped from rest from a height of 20 m above the ground. Ignore air resistance.

4.1 State the principle of conservation of mechanical energy. (2)

4.2 Calculate the gravitational potential energy of the ball at the top. (3)

4.3 Calculate the kinetic energy of the ball when it is 5 m above the ground. (4)

4.4 Calculate the speed of the ball just before it hits the ground. (4)

4.5 Calculate the kinetic energy of a 1 200 kg car moving at 20 $m \cdot s^{-1}$. (3)

4.6 By what factor does the kinetic energy of the car change if its speed doubles? (2)

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QUESTION 5

A transverse wave has a frequency of 50 Hz and a wavelength of 0,4 m.

5.1 Define a *transverse wave*. (2)

5.2 Calculate the speed of the wave. (3)

5.3 Calculate the period of the wave. (3)

5.4 Two pulses with amplitudes of 3 cm and 2 cm, both on the same side of the rest position, meet. State the principle of superposition and give the amplitude of the resulting pulse. (3)

5.5 A learner stands 170 m from a cliff, shouts, and hears the echo 1,0 s later. Calculate the speed of sound in air. (3)

5.6 Is sound a longitudinal or a transverse wave? (1)

5.7 Which property of a sound wave determines its pitch? (1)

5.8 A dog whistle produces sound with a frequency of 25 kHz. Can humans hear it? Explain. (2)

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QUESTION 6

6.1 Arrange the following in order of INCREASING frequency: X-rays, radio waves, visible light, microwaves. (2)

6.2 What is meant by a *photon*? (2)

6.3 Calculate the energy of a photon of light with a frequency of 6×10^{14} Hz. (3)

6.4 Calculate the wavelength of this light. (3)

6.5 Give ONE harmful effect of too much ultraviolet radiation on people. (1)

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QUESTION 7

Two identical metal spheres, A and B, are on insulated stands. Sphere A has a charge of +8 nC and sphere B has a charge of −4 nC. The spheres touch and are then separated.

7.1 State the principle of conservation of charge. (2)

7.2 Calculate the charge on each sphere after they are separated. (2)

7.3 Calculate the number of electrons transferred, and state the direction in which they moved. (4)

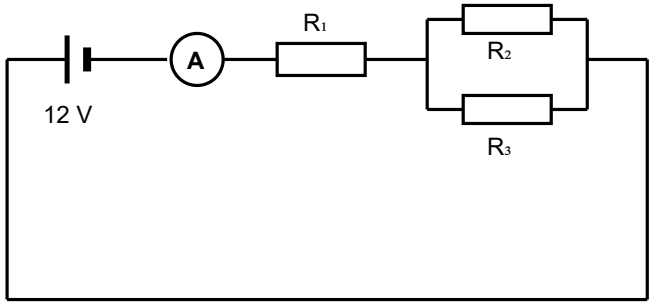
7.4 A negatively charged balloon sticks to a neutral wall. Explain why. (2)

7.5 Fuel trucks are earthed with a metal chain or strap. Explain why this is done. (2)

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QUESTION 8

In the circuit below, the battery has an emf of 12 V and negligible internal resistance. $R_1 = 4\ \Omega$, $R_2 = 6\ \Omega$ and $R_3 = 3\ \Omega$.



8.1 Define *electric current*. (2)

8.2 Calculate the effective resistance of R_2 and R_3 in parallel. (3)

8.3 Write down the total resistance of the circuit. (1)

8.4 Calculate the reading on ammeter A. (3)

8.5 Calculate the potential difference across R_1. (3)

8.6 Calculate the current through R_3. (3)

8.7 Calculate the charge that flows through the battery in 2 minutes. (3)

8.8 R_2 is now removed from the circuit. Will the reading on the ammeter INCREASE, DECREASE or STAY THE SAME? Explain. (3)

8.9 How must a voltmeter be connected to measure the potential difference across R_1? (1)

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QUESTION 9

9.1 Draw the magnetic field pattern between the north pole of one bar magnet and the south pole of another bar magnet placed close to it. Show the direction of the field lines. (3)

9.2 A compass needle points towards the Earth's geographic North Pole. Explain why, in terms of magnetic poles. (2)

9.3 Name ONE way to show that there is a magnetic field around a straight wire carrying a current. (1)

9.4 State the right-hand rule for the direction of the magnetic field around a straight current-carrying conductor. (2)

9.5 Give TWO uses of magnets or magnetism in everyday life. (2)

[10]

TOTAL: 150

SPACE FOR ROUGH WORK

END OF QUESTION PAPER